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Roll No

ES-221-CBCS

B.E. IV Semester

Examination, June 2020

Choice Based Credit System (CBCS)

System Engineering

(Common for all Branches)

Time : Three Hours

Maximum Marks : 60

- Note :** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) What is system? State some examples of systems required in system engineering. Discuss motivation and challenges of system engineering.
b) What are the elements required in work environment of system engineering discuss in detail? Also write examples of systems engineering fields.
2. a) How the system engineering method is applied in successive phase of the system life cycle, Explain?
b) What steps can the system engineer take to help ensure that system components designed by different technical groups or contractors will fit together and interact effectively when assembled to make up the total system? Discuss in terms of software system elements.
3. Draw a context diagram for a standard washing machine. Make sure to identify all of the external entities and label all of the interactions.

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4. Describe the characteristics of a set of well stated operational requirements. State what could be the result if a requirement did not have these characteristics.
5.
 - a) Describe functional analysis and design in detail.
 - b) Describe prototype development model in detail.
6. What is in compatible operational requirements? Discuss the concept exploration for “new aircraft”.
7. Describe what is meant by the term functional allocation illustrate its application in software engineering.

OR

What is Software engineering management? How the transition from development to production is done explain.

8. Write short notes on any two:
 - a) Risk mitigation technique
 - b) Operational test and evaluation
 - c) Complexity of modern systems
